

Purpose of This Paper

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The gap this paper fills

Almost every method for approximating a function comes with an error term, and almost every one of those error terms is controlled using Taylor's theorem. The version with the Lagrange remainder says that if you approximate a function by a polynomial of degree n , the error is exactly

$$\frac{f^{(n+1)}(\xi)}{(n+1)!} (b-a)^{n+1}$$

for some point ξ somewhere inside the interval.

That statement is strange when you look at it directly. It is an equality, not an estimate, so it is telling us something exact. But the exactness is useless on its own, because the theorem refuses to say which point ξ is. It says only that one exists. Every textbook proves the existence and then immediately gives up on the location, replacing $f^{(n+1)}(\xi)$ by the largest value the derivative takes anywhere on the interval. That replacement is where the sharpness is lost, and it is lost because nobody knows where ξ is.

This paper asks the question the textbooks skip. Given reasonable information about the function, how much can we actually say about the location of ξ ?

Why this is a continuation and not a new project

When $n = 0$, Taylor's theorem is the mean value theorem. My previous paper found a bound on the location of the mean value point, placing it near the balance point of a certain weight.

At the time I thought of that as a self-contained problem. It was not. It was the first case of an infinite family, one case for each order n , and I had solved it without realizing that was what I was doing. This paper takes the same machinery, namely the weight, the balance point and the measure of spread, and carries it up to every order.

That structure gives the project a built-in correctness check. Setting $n = 0$ in the new result must reproduce the old result exactly. If it did not, the general form would be wrong. It does, and that check is Figure 3.

What the paper actually establishes

The paper turns on one observation, which took much longer to find than to state. Written in normalized coordinates, Taylor's theorem says precisely this:

The intermediate point is the place where the $(n + 1)$ st derivative equals its own weighted average, taken against a specific weight determined by the order n .

Once the problem is phrased that way, it stops being a question about functions and becomes a question about a single probability weight. Everything else follows. Four things come out of it.

The classical answer becomes exact. It has been known since 1982 that as the interval shrinks to a point, the position of ξ tends to the fraction $1/(n + 2)$ of the way across. In the new framing, $1/(n + 2)$ is simply the balance point of the weight, and it is the exact answer on any interval, of any length, whenever the relevant derivative is a straight line. No shrinking is required. The old result is recovered as a special case of something cleaner.

The region is described completely. The main result does not give a bound on where ξ can be. It gives the exact set of positions ξ can occupy, together with the specific functions that put it at each end of that set. Because the description is complete, there is no gap left between what is proved and what is true.

One hypothesis turns out to be doing all the work. If the next derivative after the one in the remainder is allowed to change sign anywhere on the interval, then ξ can be placed anywhere at all and no bound of any kind exists. This is proved, not assumed. It means the whole subject is really a statement about strict monotonicity, and it draws a sharp line around where any result of this type can apply.

The headline is the opposite of what I expected. I began the project expecting that higher order approximations would pin down their remainder point more tightly, since better approximations ought to know more. In absolute terms this is true, and the region shrinks at the rate $1/n$. But the anchor point shrinks at exactly the same rate, so the honest comparison is the relative one, and there the region does not tighten. It widens, steadily, approaching a fixed limiting shape. Measured against its own distance from the base point, the intermediate point becomes slightly freer as the approximation improves, not more constrained.

That last finding is the one I would not have predicted, and it is the reason the project was worth doing rather than merely worth planning.

How to read what follows

The paper is arranged so that the argument is visible before the machinery.

Section 2 contains the reduction, which is the only genuinely new idea and is half a page long. Section 3 contains the main result. A reader who wants the substance and nothing else can stop after Section 3.

Sections 4 through 6 are consequences and extensions: explicit formulas for small orders, the proof that the sign condition cannot be dropped, and the version with general weights. Section 7 contains the finding about growing order described above. Section 8 reports the numerical checks, which were run before each statement was proved rather than after.

Every claim in the paper was tested numerically first. Where the numbers disagreed with what I expected, I changed what I expected. Two of the four predictions I wrote down at the start of the project turned out to be wrong in their details, and both are corrected in the text rather than quietly dropped.

What is not settled

One conjecture is stated and not proved. The relative widening described above is confirmed numerically across a wide range and proved only in the limiting case of small distortion. A full proof appears to need a monotonicity property I was unable to establish, and I have stated it as an open question rather than claiming more than I have.

Four further open questions are listed at the end of the paper. The most promising, in my judgment, is the practical one: since the paper produces an interval for ξ rather than a single worst case value, it produces a two sided estimate of the approximation error where the standard treatment gives only a one sided one. Measuring how much that actually gains in real error bounds is a concrete next project.